

Arun Kumar Thiyagarajan

Aspiring AI & Machine Learning Engineer

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PROFESSIONAL SUMMARY

Innovative AI & Machine Learning Engineer with hands-on experience developing real-world AI applications using Python, Machine Learning, and Deep Learning. Skilled in building intelligent automation systems, chatbots, and predictive solutions. Strong analytical, problem-solving, and teamwork abilities with a focus on delivering practical, scalable AI solutions.

EDUCATION

United Institute Of Technology 11/2022 – 05/2026
Bachelor of Technology (B.Tech) in Artificial Intelligence and Data Science

IIT(Patna) 02/2026 – 06/2026
Certification program in Artificial Intelligence and Machine Learning Essentials

SKILLS

— **Programming Languages:** Python, SQL | **Data Preparation & Analysis:** Data Cleaning, Exploratory Data Analysis (EDA), Feature Engineering, Data Visualization | | **ML Modeling & Evaluation:** Regression, Classification, Model Training, Model Evaluation (Accuracy, Precision, Recall, F1-Score) | | **Frameworks & Libraries:** Scikit-learn, Pandas, NumPy, Matplotlib | | **Tools & Workflow:** Git, GitHub, Jupyter Notebook, Google Colab, Visual Studio Code | **Soft Skills:** Stakeholder communication, Team collaboration, Analytical thinking.

PROJECTS

AI PDF QUESTION ANSWERING SYSTEM | Python | Streamlit | LangChain | FAISS | Google Gemini

- Developed an AI-powered application that answers user questions based on uploaded PDF documents using a Retrieval-Augmented Generation (RAG) pipeline.
- Extracted PDF content using PyPDF, split documents into smaller chunks, and generated vector embeddings for efficient document retrieval.
- Executed FAISS vector search with LangChain to retrieve the most relevant document sections before generating responses.
- Merged Google Gemini to generate context-aware answers based only on the retrieved PDF content.
- Built an interactive Streamlit interface that enables users to upload PDF files and ask questions in real time.

AI RESUME PARSER | Python | Streamlit | Google Gemini | PyPDF

- Expanded an AI-powered Resume Parser that extracts structured information from PDF resumes using Large Language Models.
- Built an interactive Streamlit interface to upload resumes and automatically process PDF content using PyPDF.
- Blended Google Gemini API to identify and extract candidate details such as personal information, education, skills, experience, and projects into structured JSON format.
- Enforced environment variable management using dotenv for secure API key handling and improved application security.
- Designed a clean and user-friendly workflow for automated resume screening, reducing manual data extraction effort.

Bank Customer Churn Prediction | Machine Learning

- Improved a machine learning-based web application to predict whether a bank customer is likely to churn based on customer demographics and account details.
- Performed data preprocessing, feature encoding, and exploratory data analysis (EDA) on customer data.
- Trained and evaluated multiple machine learning models to classify customer churn and selected the best-performing model.
- Built an interactive web application using Flask to collect customer information and display real-time churn predictions.
- Executed categorical feature encoding and integrated the trained model using Joblib for deployment.
- **Tech Stack:** Python, Pandas, NumPy, Scikit-learn, Flask, HTML, CSS, Joblib

CERTIFICATES

Artificial Intelligence & Data Scientist — Tamil Nadu Academy of Construction | Completed: 2026

Applied Artificial Intelligence and Machine Learning Essentials — IIT Patna, Masai School

Introduction to Generative AI — Google Cloud | Completed: 2025

IBM Virtual Internship via FSP – Web, Mobile Development & Marketing — Successfully completed IBM Virtual Internship via FSP (Feb 5 – Mar 5, 2026), gaining foundational knowledge in Web Development, Mobile Application Development, and Digital Marketing through the IBM Developer Skills Network.